READING GROUP

MedFact: Benchmarking the Fact-Checking Capabilities of LLMs on Chinese Medical Texts

A diverse, uncontaminated benchmark for safe medical AI

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Medical LLMs need rigorous fact-checking

- LLMs increasingly draft and answer **medical content**
- A single factual error can **harm patients**
- Deployment demands **regulatory compliance**, not just fluency
- Existing benchmarks test QA, not **error detection in real text**
- We need to know: can models **catch medical errors**?

CONTRIBUTIONS

What this paper delivers

- **MedFact**: 2,116 expert-annotated Chinese medical texts
- Diverse and **contamination-resistant** by design
- Evaluation of **20 leading LLMs** on two tasks
- Models trail **human experts**, especially on localization
- We expose an **'over-criticism' failure mode**

A hybrid AI-human construction pipeline

- Start from **27,116 private texts** (no contamination)
- **7-LLM filter** by majority vote, Fleiss κ 0.75
- **3 expert-in-the-loop** rounds → 96.4% agreement
- Filtered to **6,405**, then hard-case mining + augmentation
- Final review → **2,116 balanced** instances

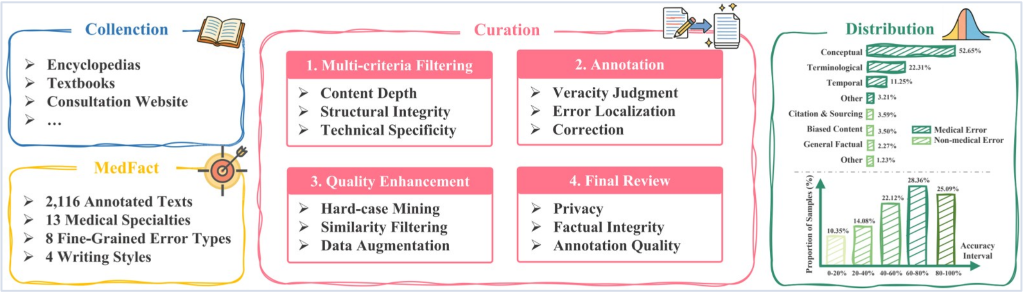

Figure 1: Data construction pipeline and MedFact components.

DIVERSITY

Broad, realistic coverage

- **13 medical specialties** across 4 high-level domains
- Led by **Internal Medicine** (~55%), then OB-GYN, Surgery
- **4 writing styles**: encyclopedia, journalism, Q&A, misinformation
- **5 difficulty levels** by model success rate
- Mirrors **real-world** medical information

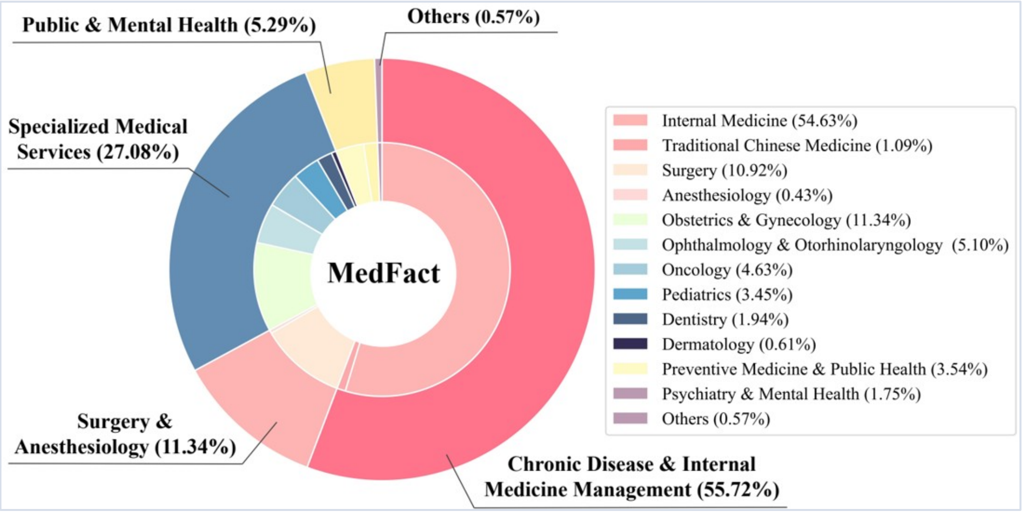

Figure 3: Distribution of specialties in MedFact.

Errors are overwhelmingly medical

- **89.4% medical** errors vs 10.6% non-medical
- **Conceptual 52.65%** dominates the set
- Terminological **22.31%**, Temporal 11.25%
- Non-medical: citation, bias, **general factual**
- Catching errors requires **deep medical knowledge**

Error Category	Count	Proportion (%)
I. Medical Errors		
Conceptual	557	52.65
Terminological	236	22.31
Temporal	119	11.25
Other Medical	34	3.21
II. Non-Medical Errors		
Citation & Sourcing	38	3.59
Biased Content	37	3.50
General Factual	24	2.27
Other Non-Medical	13	1.23

Table 1: Distribution of 8 fine-grained error types (1,058 incorrect instances).

Two tasks: detect, then localize

- **Veracity Classification (VC)**
  - Binary: is the text correct or incorrect?
  - 'Incorrect' is the positive class
  - The **easier** task
- Metrics: **Precision, Recall, F1**

- **Error Localization (EL)**
  - Pinpoint the exact error span
  - Only on texts flagged incorrect
  - The **stringent** test of knowledge
- Judged by **GPT-4o** ($\kappa \approx 0.87$ vs experts)

Models detect errors, but can't localize them

- Proprietary models **beat open-source** throughout
- Best EL F1 **0.686** (XiaoYi+CoT) < human **0.701**
- Every model scores **lower on EL than VC**
- 'Correct-for-the-wrong-reason': **right label, wrong span**
- EL exposes **medical-knowledge deficits**

Full results: 20 LLMs, zero-shot and CoT

Model	Zero-shot				CoT			
	VC	EL			VC	EL		
	F1	Precision	Recall	F1	F1	Precision	Recall	F1
Human	0.7521	0.7495	0.6588	0.7012	–	–	–	–
Open-source								
II-Medical-8B	0.6361	0.3278	0.2618	0.2911	0.6477	0.2904	0.2240	0.2529
HuatuoGPT-o1-7B	0.5831	0.3054	0.1654	0.2146	0.5932	0.3508	0.1900	0.2465
DeepSeek-R1	0.6847	0.5035	0.7580	0.6051	0.6764	0.5012	0.7826	0.6111
DeepSeek-V3	0.6444	0.5832	0.3809	0.4608	0.6724	0.5266	0.5510	0.5386
Qwen2.5-72B-Instruct	0.5195	**0.8024**	0.3223	0.4599	0.6400	<u>0.6731</u>	0.4282	0.5234
Qwen3-235B-A22B	0.6944	0.6234	0.5803	0.6011	0.6468	0.6565	0.5076	0.5725
DeepSeek-R1-Distill-Llama-70B	0.6774	0.4990	0.7410	0.5964	0.6882	0.5142	0.7873	0.6221
DeepSeek-R1-Distill-Qwen-32B	0.5524	0.6648	0.3450	0.4543	0.5718	0.6480	0.3393	0.4454
Proprietary								
XiaoYi	<u>0.7126</u>	0.6512	0.7023	**0.6758**	**0.7061**	0.6530	0.7221	**0.6858**
GPT-4o	0.6741	0.4966	0.5595	0.5262	0.6635	0.5012	0.6144	0.5520
GPT-4.5	0.6965	0.5694	0.6824	0.6208	0.6952	0.5971	0.6947	0.6422
o1	0.6713	<u>0.6921</u>	0.5652	0.6223	0.6810	**0.6975**	0.5775	0.6319
o3	0.6890	0.5355	<u>0.8129</u>	0.6456	0.6870	0.5579	<u>0.8658</u>	0.6785
Claude 3.7 Sonnet	0.6900	0.5242	0.5728	0.5474	0.6943	0.5567	0.6777	0.6113
Gemini 2.5 Flash	0.6787	0.4988	0.7921	0.6121	0.6730	0.4927	0.8299	0.6183
Gemini 2.5 Pro	0.6658	0.4689	**0.8346**	0.6005	0.6667	0.4828	**0.8752**	0.6223
Grok-4	0.6951	0.2675	0.2316	0.2482	0.6908	0.2136	0.1692	0.1888
Qwen2.5-Max	0.7006	0.6113	0.5113	0.5569	0.6942	0.5663	0.6219	0.5928
ERNIE-X1	0.6792	0.5580	0.5775	0.5676	0.6871	0.6240	0.5945	0.6089
Doubao-Seed-1.6	0.7122	0.5566	0.6928	0.6173	0.7006	0.5609	0.7268	0.6332
Doubao-Seed-1.6-thinking	**0.7139**	0.6501	0.6938	<u>0.6712</u>	<u>0.7050</u>	0.6307	0.7344	<u>0.6786</u>

Table 3: Full MedFact results (zero-shot & CoT). Best bold, 2nd-best underlined; human = avg of 3 professionals.

Failures are knowledge failures

- **~76% of DeepSeek-R1 errors**: insufficient medical knowledge
- **Concept conflation** of related drugs or conditions
- **Outdated information**: superseded guidelines look plausible
- **Logical inconsistency**: rationale contradicts verdict
- Fixes: knowledge currency, **discrimination, consistency**

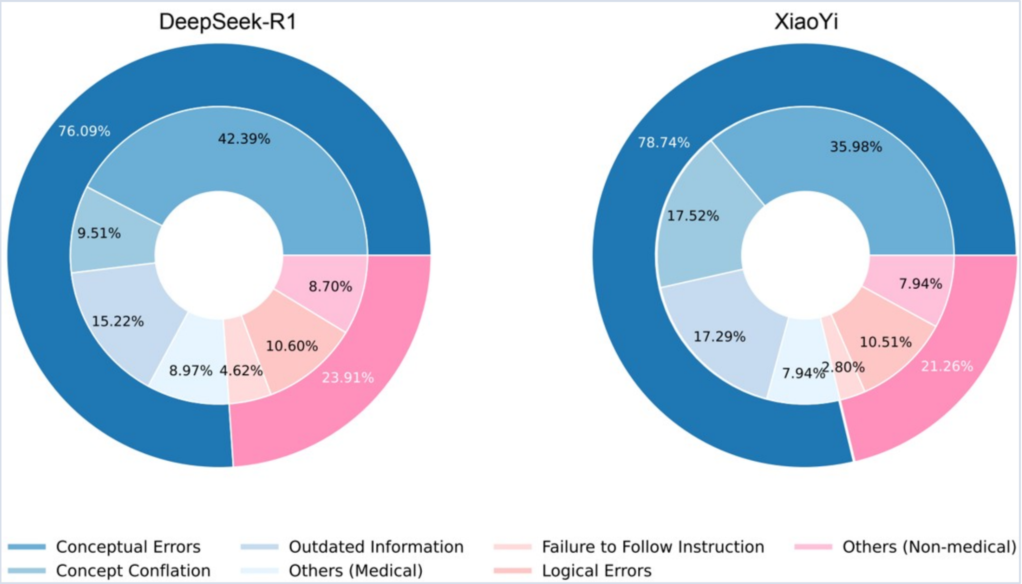

Figure 4: Error distribution of DeepSeek-R1 and XiaoYi on the EL task (zero-shot).

OVER-CRITICISM

Advanced reasoning makes models over-critical

- **RAG helps** both tasks with relevant sources
- MAD & MDAgents: **recall up, precision down**
- Same **'precision down, recall up'** across all 5 models
- Agents reward **flagging novel errors**, not concurrence
- **Budget-forcing** gives no gain, hallucinates errors

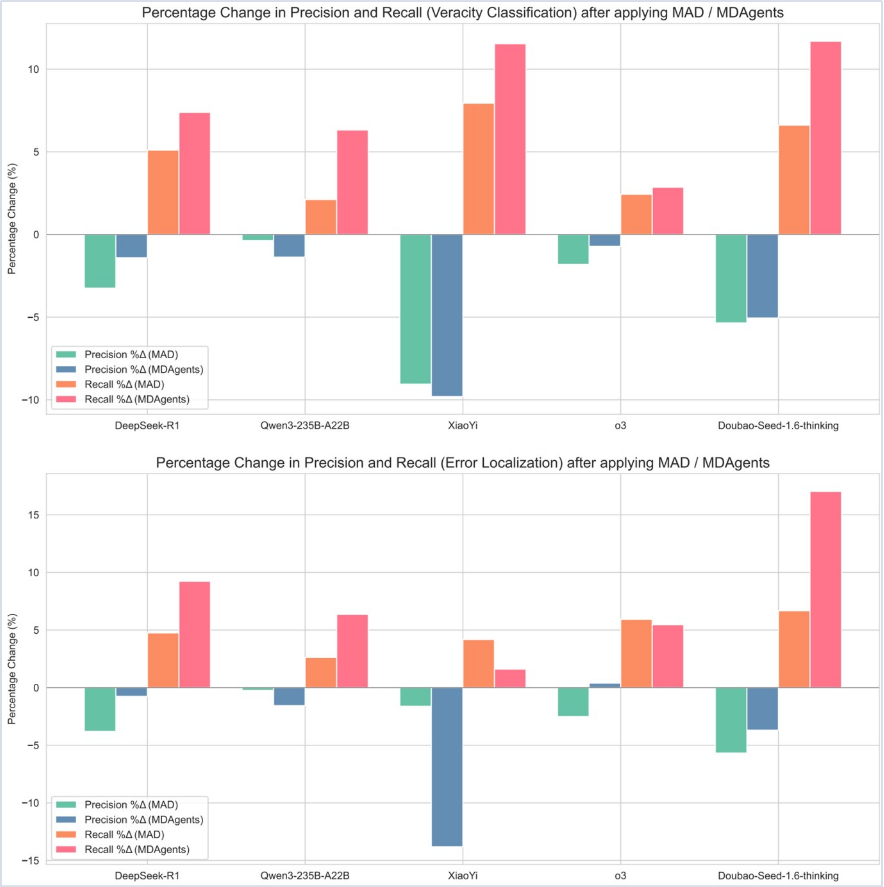

Figure 8: Precision/recall changes after applying MAD and MDAgents.

Over-criticism persists across languages

- Tested on an **English-translated MedFact**
- Both models gain **F1** (Gemini 0.622 → 0.674)
- Recall rises; **precision can drop** (Doubao)
- Models flag errors **more aggressively in English**
- Language shifts the **precision-recall trade-off**

TAKEAWAYS

What to remember

- **MedFact**: 2,116 expert texts, diverse and uncontaminated
- Models **detect** errors but **can't localize** them like humans
- Failures are **medical-knowledge** failures, not reasoning style
- **Over-criticism** worsens with multi-agent & inference scaling
- For safety, **prioritize knowledge** and balanced precision-recall

Scope and responsible release

- **Chinese-only**, tied to China's healthcare system
- EL uses a **GPT-4o judge** (scalable, but imperfect)
- Medical knowledge is **fixed at creation time**
- IRB-approved; **all PII removed**; fair annotator pay
- **Research-only** license: no clinical or commercial use